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#soal nomor 5
import math

# Fungsi untuk menghitung jarak Euclidean 3 dimensi
def hitung_jarak(titik1, titik2):
    return math.sqrt(
        (titik1[0] - titik2[0]) ** 2 +
        (titik1[1] - titik2[1]) ** 2 +
        (titik1[2] - titik2[2]) ** 2
    )

# Fungsi untuk menentukan cluster
def tentukan_cluster(titik_u):
    cluster_a = (2, 1, 3)
    cluster_b = (1, -4, 6)
    cluster_c = (-2, 3, -2)

    jarak_a = hitung_jarak(titik_u, cluster_a)
    jarak_b = hitung_jarak(titik_u, cluster_b)
    jarak_c = hitung_jarak(titik_u, cluster_c)

    print(f"Jarak ke Cluster A = {jarak_a}")
    print(f"Jarak ke Cluster B = {jarak_b}")
    print(f"Jarak ke Cluster C = {jarak_c}")

    if jarak_a < jarak_b and jarak_a < jarak_c:
        return "A"
    elif jarak_b < jarak_a and jarak_b < jarak_c:
        return "B"
    else:
        return "C"

# Program utama
x1 = float(input("Masukkan x1: "))
x2 = float(input("Masukkan x2: "))
x3 = float(input("Masukkan x3: "))

titik_u = (x1, x2, x3)

hasil_cluster = tentukan_cluster(titik_u)

print("Titik U termasuk Cluster", hasil_cluster)
```

```
Masukkan x1: 1
Masukkan x2: 4
Masukkan x3: 3
Jarak ke Cluster A = 3.1622776601683795
Jarak ke Cluster B = 8.54400374531753
Jarak ke Cluster C = 5.916079783099616
Titik U termasuk Cluster A
```

```
#Soal nomor 5 (tidak valid)
import math

def hitung_jarak(titik1, titik2):
    return math.sqrt(
        (titik1[0] - titik2[0]) ** 2 +
        (titik1[1] - titik2[1]) ** 2 +
        (titik1[2] - titik2[2]) ** 2
    )

def tentukan_cluster(titik_u):
    cluster_a = (2, 1, 3)
    cluster_b = (1, -4, 6)
    cluster_c = (-2, 3, -2)

    jarak_a = hitung_jarak(titik_u, cluster_a)
    jarak_b = hitung_jarak(titik_u, cluster_b)
```

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    jarak_c = hitung_jarak(titik_u, cluster_c)

    print(f"Jarak ke Cluster A = {jarak_a}")
    print(f"Jarak ke Cluster B = {jarak_b}")
    print(f"Jarak ke Cluster C = {jarak_c}")

    if jarak_a <= jarak_b and jarak_a <= jarak_c:
        return "A"
    elif jarak_b <= jarak_a and jarak_b <= jarak_c:
        return "B"
    else:
        return "C"

try:
    x1 = float(input("Masukkan x1: "))
    x2 = float(input("Masukkan x2: "))
    x3 = float(input("Masukkan x3: "))

    titik_u = (x1, x2, x3)

    hasil = tentukan_cluster(titik_u)

    print("Titik U termasuk Cluster", hasil)

except ValueError:
    print("Input harus berupa angka!")

```

```

Masukkan x1: 4
Masukkan x2: -6
Masukkan x3: 3
Jarak ke Cluster A = 7.280109889280518
Jarak ke Cluster B = 4.69041575982343
Jarak ke Cluster C = 11.916375287812984
Titik U termasuk Cluster B

```

```

#Soal nomor 5 (tidak valid)
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def hitung_jarak(titik1, titik2):
    return math.sqrt(
        (titik1[0] - titik2[0]) ** 2 +
        (titik1[1] - titik2[1]) ** 2 +
        (titik1[2] - titik2[2]) ** 2
    )

def tentukan_cluster(titik_u):
    cluster_a = (2, 1, 3)
    cluster_b = (1, -4, 6)
    cluster_c = (-2, 3, -2)

    jarak_a = hitung_jarak(titik_u, cluster_a)
    jarak_b = hitung_jarak(titik_u, cluster_b)
    jarak_c = hitung_jarak(titik_u, cluster_c)

    print(f"Jarak ke Cluster A = {jarak_a}")
    print(f"Jarak ke Cluster B = {jarak_b}")
    print(f"Jarak ke Cluster C = {jarak_c}")

    if jarak_a <= jarak_b and jarak_a <= jarak_c:
        return "A"
    elif jarak_b <= jarak_a and jarak_b <= jarak_c:
        return "B"
    else:
        return "C"

try:
    x1 = float(input("Masukkan x1: "))
    x2 = float(input("Masukkan x2: "))
    x3 = float(input("Masukkan x3: "))

    titik_u = (x1, x2, x3)

```

```

hasil = tentukan_cluster(titik_u)

print("Titik U termasuk Cluster", hasil)

except ValueError:
    print("Input harus berupa angka!")

```

```

Masukkan x1: -4
Masukkan x2: 3
Masukkan x3: -3
Jarak ke Cluster A = 8.717797887081348
Jarak ke Cluster B = 12.449899597988733
Jarak ke Cluster C = 2.23606797749979
Titik U termasuk Cluster C

```

```

#Soal nomor 5 (tidak valid)
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def hitung_jarak(titik1, titik2):
    return math.sqrt(
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        (titik1[1] - titik2[1]) ** 2 +
        (titik1[2] - titik2[2]) ** 2
    )

def tentukan_cluster(titik_u):
    cluster_a = (2, 1, 3)
    cluster_b = (1, -4, 6)
    cluster_c = (-2, 3, -2)

    jarak_a = hitung_jarak(titik_u, cluster_a)
    jarak_b = hitung_jarak(titik_u, cluster_b)
    jarak_c = hitung_jarak(titik_u, cluster_c)

    print(f"Jarak ke Cluster A = {jarak_a}")
    print(f"Jarak ke Cluster B = {jarak_b}")
    print(f"Jarak ke Cluster C = {jarak_c}")

    if jarak_a <= jarak_b and jarak_a <= jarak_c:
        return "A"
    elif jarak_b <= jarak_a and jarak_b <= jarak_c:
        return "B"
    else:
        return "C"

try:
    x1 = float(input("Masukkan x1: "))
    x2 = float(input("Masukkan x2: "))
    x3 = float(input("Masukkan x3: "))

    titik_u = (x1, x2, x3)

    hasil = tentukan_cluster(titik_u)

    print("Titik U termasuk Cluster", hasil)

except ValueError:
    print("Input harus berupa angka!")

```

```

Masukkan x1: 9
Masukkan x2: 4
Masukkan x3: a
Input harus berupa angka!

```

